

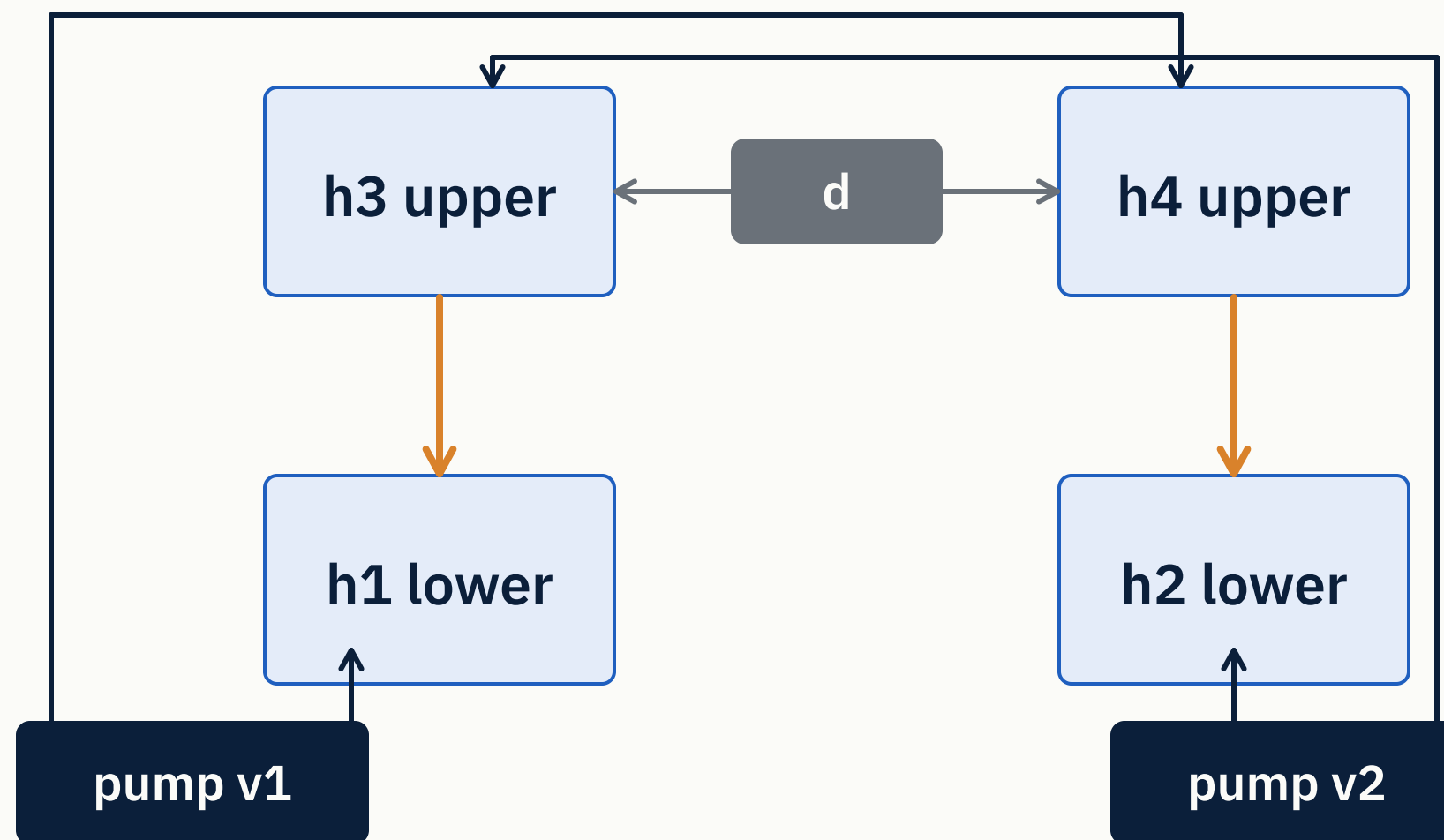
Which causal-discovery algorithm recovers the quadruple-tank wiring from data alone?

PC, PCMCI+ and VAR-LiNGAM tested on a simulated four-tank process with a known answer key: 7 signals, 8 true links, two operating points.

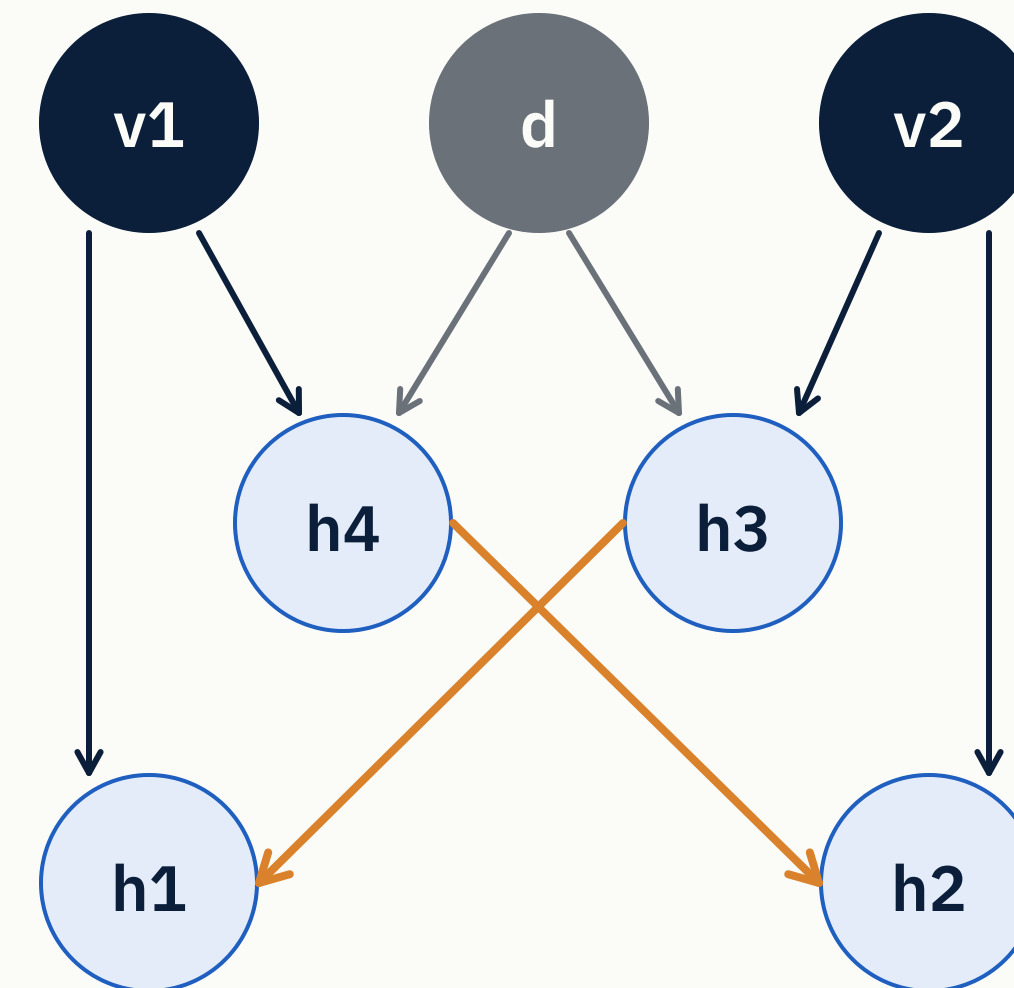
THE SYSTEM

Two pumps, four tanks, eight pipes: the graph on the right is the answer key for every result that follows

Physical rig (Johansson quadruple-tank process)



Same system as a causal graph: 7 nodes, 8 arrows



Pumps split between a lower and an upper tank; upper tanks drain into the tank below (**orange** = the two drainage links, $h3 \rightarrow h1$ and $h4 \rightarrow h2$); d feeds both upper tanks.

THE DATASETS

Two datasets because the same rig behaves two different ways depending on how the valves split the flow

Same 8 true arrows and same inputs in both: 4 h at 5 s sampling, 2880 rows each. Only the strength and timing of each arrow changes.

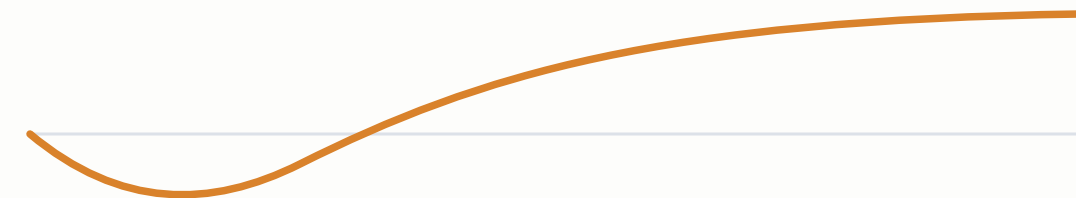
P- · minimum phase · the "easy" control case



Lower-tank level after a pump step: rises promptly

- Most of each pump's flow goes straight to the lower tanks
- Drainage links carry the smaller share of water
- Time constants 23–91 s

P+ · non-minimum phase · the "hard" control case



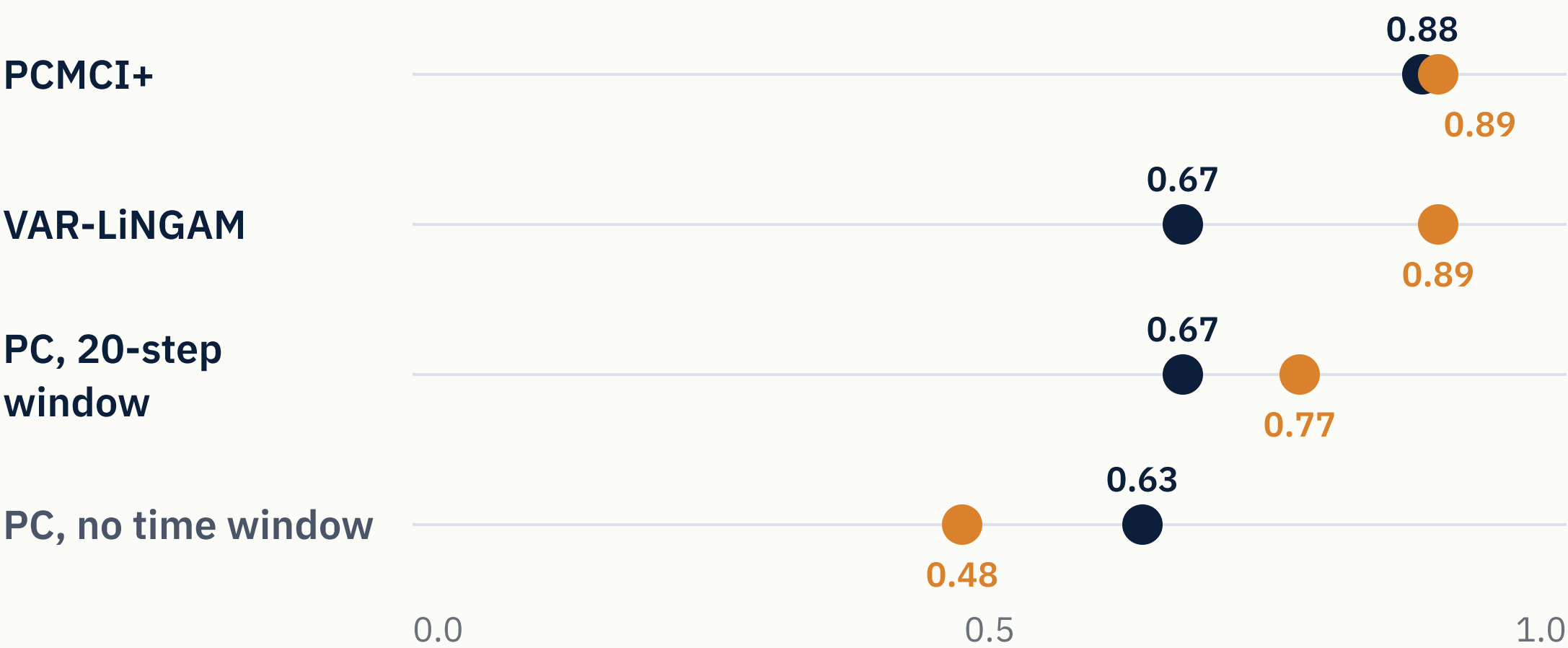
Lower-tank level after a pump step: briefly moves the wrong way, then follows

- Most of each pump's flow goes to the upper tanks first
- Drainage links carry the larger share of water
- Time constants 39–91 s

EXECUTIVE SUMMARY

PCMCI+ recovers the wiring best at both operating points; the two drainage links cause almost every error

F1 score on the 8 true links (1.0 = perfect). ● P- ● P+



2 of 8
edges wrong for PCMCI+ at either operating point (SHD 2)

1.5 s vs 134 s
runtime per dataset, PCMCI+ vs VAR-LiNGAM

2 arrows
the tank-to-tank drainage links (h3→h1, h4→h2) account for nearly every miss or flip

HOW WE SCORE

Every algorithm hands back a graph; we count how many of the 8 true arrows it found, missed, flipped or invented

THE 7 SIGNALS

- v** v1, v2: pump voltages. Driven by us, square waves
- d** d: disturbance flow into the upper tanks
- h** h1–h4: tank levels. The outputs the algorithms must explain

All three inputs are recorded, so nothing hidden drives two tanks at once.

THREE SCORES

Precision: of the arrows reported, how many are real

Recall: of the 8 real arrows, how many were reported, pointing the right way

F1: the balance of the two. 1.0 means a perfect graph

SHD: number of edits (add, delete, flip) to reach the true graph. Lower is better

Lagged results are collapsed to one arrow per pair before scoring. Self-loops are ignored.

SHARED SETTINGS

Significance: 0.01

Time window: 20 steps (100 s), about one time constant of the slowest tank

Test: partial correlation (linear)

VAR-LINGAM: 100 bootstrap resamples, keep a link only if present in 90 of them

Row count used by the main run is not recorded in its summary. UNVERIFIED whether it used all 2880 rows.

THE METHODS

Three ways of asking "does A drive B?", each with a different blind spot

PC

Removes a link whenever two signals become unrelated once you control for the others.

Built for snapshots, not time series. We ran it twice: with no time window, and with the same 20-step window as the others.

Blind spot: no correction for signals that mostly predict themselves

PCMCI+

PC redesigned for time series. Tests each link at every lag in the window and corrects for autocorrelation.

The method the literature recommends for slow, autocorrelated process data. Our pre-registered favourite.

Blind spot: linear test only, in this run

VAR-LINGAM

Fits a linear forecasting model, then uses the shape of the noise (non-Gaussian) to decide which way each link points.

Keeps a link only if it survives 90 of 100 bootstrap resamples.

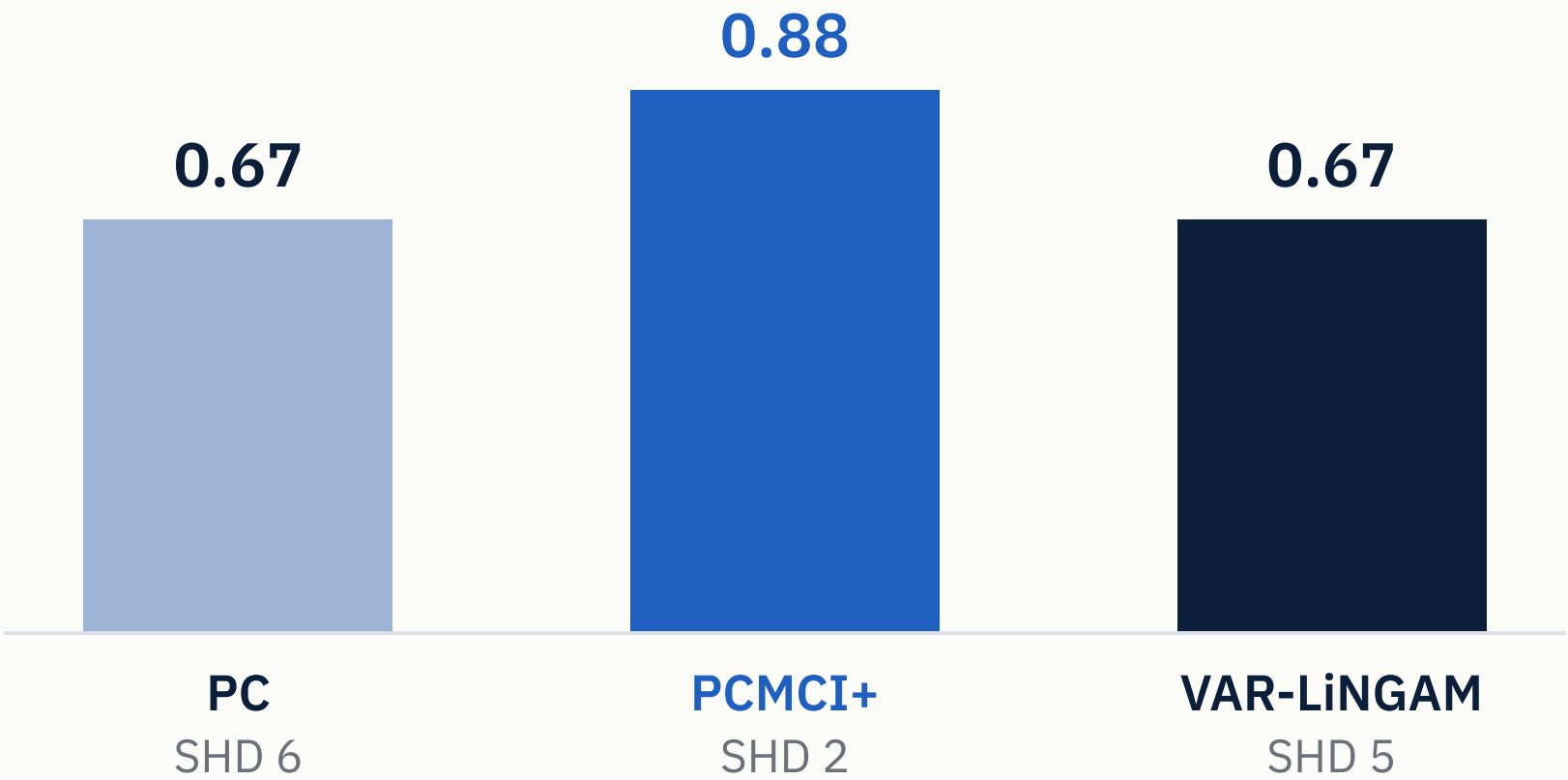
Blind spot: direction is unreliable where the noise is Gaussian (the tank levels)

HEADLINE RESULT

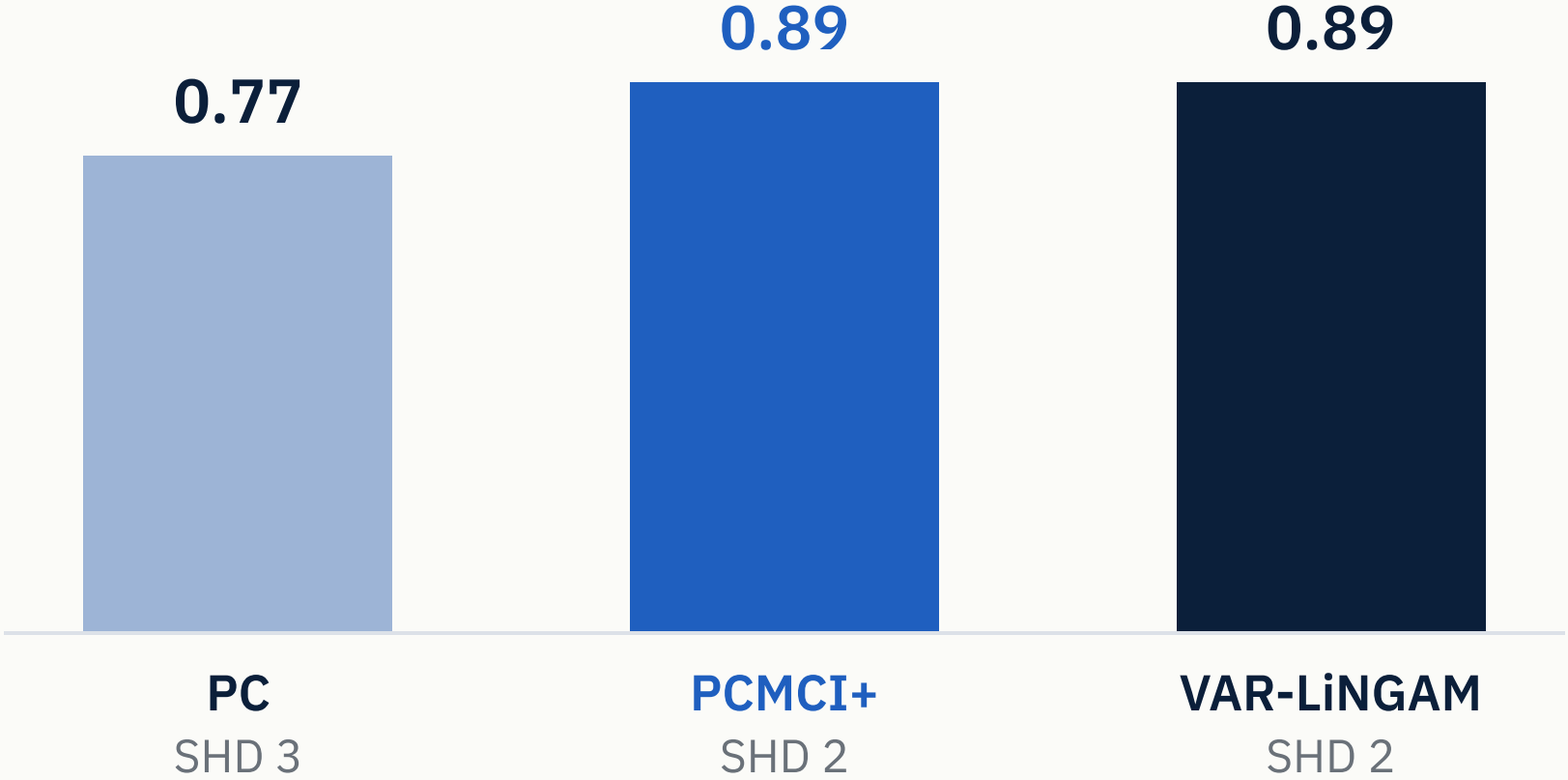
PCMCI+ recovers 7 to 8 of the 8 true links with at most 2 mistakes at either operating point

F1 score on 8 directed edges; SHD (edits to reach the true graph) shown under each bar. 20-step window, partial correlation, significance 0.01.

P- · minimum phase



P+ · non-minimum phase



PCMCI+ is the only method whose worst case stays above F1 0.85. Every true link it reports points the right way: no reversed edges in either run.

The two tank-to-tank drainage links are what separate the methods; the six input links are found by everyone

Result for each orange arrow from slide 2, at P- and P+

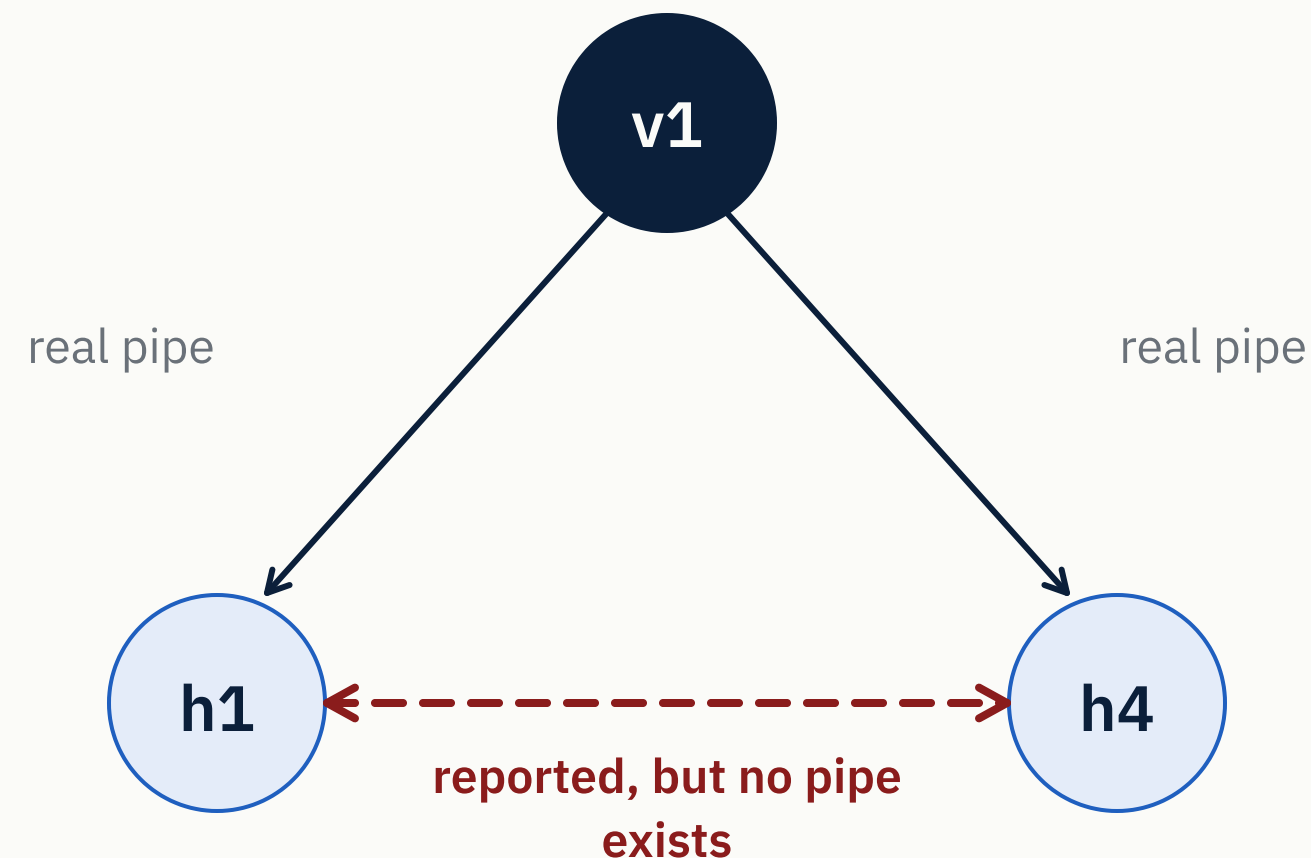
LINK	PC (windowed)	PCMCI+	VAR-LiNGAM
h3 → h1 upper tank 3 drains into lower tank 1	P- missed P+ missed	P- missed P+ found	P- reversed (h1→h3) P+ found
h4 → h2 upper tank 4 drains into lower tank 2	P- missed P+ missed	P- found P+ found	P- missed P+ found

These are the slow, weak links. An upper tank's level is almost fully explained by its own history, which hides its effect on the tank below. Windowed PC also misses one pump link (v2→h2) at both operating points.

THE RECURRING FALSE POSITIVE

Every method invents a link between the two tanks that share a pump

Pump v1 feeds both h1 and h4. Both tanks move together whenever the pump changes, and a 100 s window does not fully separate that from a direct pipe.



5/6

runs report a spurious h1–h4 link, in one direction or the other

3/6

runs report a spurious v2→h1 (the other pump reaching across)

The same pattern holds for pump v2 and its tanks h2, h3 (spurious h3→h2 or h1→h2 in the earlier run).

Practical reading: a reported link between two signals fed by the same actuator is suspect until checked against the P&ID.

BEFORE AND AFTER

Giving plain PC a time window fixes its precision but not its recall

The one methodological change between the two runs: PC scored on the current sample only, then on a 20-step window like the other methods.

PC, no time window			
	Precision	Recall	F1
P−	0.55	0.75	0.63
P+	0.38	0.63	0.48
Invented links: 4 at P−, 6 at P+. Dense, spurious, as pre-registered.			

PC, 20-step window			
	Precision	Recall	F1
P−	0.71	0.63	0.67
P+	1.00	0.63	0.77
Invented links: 2 at P−, 0 at P+. But recall stays at 0.63: it still misses both drainage links and one pump link.			

0.38 → 1.00 precision at P+. The window removes the invented links; it does not find the missing ones.

THE SECOND OPINION

VAR-LiNGAM matches PCMCI+ at P+ but pays about 90× the runtime and stumbles on direction at P−

1.5 s

PCMCI+, per dataset

At P+: all 8 true links, 2 invented (F1 0.89)

Ties PCMCI+ on F1 and SHD. The two extras are cross-pump links ($v1 \rightarrow h2$, $v2 \rightarrow h1$).

At P−: one link reversed, one missed (F1 0.67)

Reports $h1 \rightarrow h3$ instead of $h3 \rightarrow h1$ and drops $h4 \rightarrow h2$. Both are drainage links.

134 s

VAR-LiNGAM, per dataset (295–301 s in the earlier run)

Why direction fails on the tanks

Its direction rule needs non-Gaussian noise. The pump signals have it; the tank levels are near-Gaussian because the tanks smooth everything. Predicted before the run.

Where the time goes

100 bootstrap fits with 20 lags each. Not inherent to the estimator; a cheaper setting would cut it.

PREDICTIONS VS OUTCOME

The results matched the ranking we wrote down before running anything, with one surprise

Expectations recorded on 7 September 2026, unchanged since. Results from 28 September 2026.

METHOD	PREDICTED	OBSERVED
PCMCI+	Best of the set; most or all 8 links	✓ Best at both operating points, 7 and 8 of 8 links
VAR-LiNGAM	Good on lagged links, weak on tank-to-tank direction	✓ Matches at P− (one flip, one miss); better than expected at P+
PC, no window	Poor, dense, spurious	✓ F1 0.48–0.63 with 4–6 invented links



Surprise: we expected P+ to be harder and to flip orientations. Instead P+ was easier for every method, and windowed PC at P+ was the only run with zero invented links.

IMPLICATIONS

What this means for root-cause detection on real process data

1

Use PCMCI+ as the default

Best accuracy at both operating points, seconds to run, and it never reversed a link in any run.

2

Distrust links between two signals on the same actuator

Shared-pump tanks produced a spurious link in 5 of 6 runs. Check such links against the plant drawing before acting.

3

Expect slow, weak couplings to go missing

Tank-to-tank drainage was the hard case every time. Where such links are known, feed them in as prior knowledge.

Confidence limits: one simulator, two operating points, one random seed, linear tests only. No error bars exist on any number in this deck.

NEXT STEPS

Two cheap runs would let us say "robust" on the previous slide

Both are already scoped; no new algorithm code is needed.

What	Cost	What it settles
Run the planned nonlinear-test (CMIknn) comparison for PC and PCMCI+ on both operating points	About 2.7 h compute, config written on 27 Sep 2026	Whether the PCMCI+ win survives a nonlinear independence test
Repeat all runs across several noise seeds	Minutes per seed for PC and PCMCI+; about 2 min per seed for VAR-LiNGAM	Error bars on every F1 in this deck
Record the row count and stride in every run summary	One line of logging	Removes the UNVERIFIED note on slide 5